

Ranking Inversion in Emergency UAV Planning: Why the Best Navigator Is Not the Best Rescuer

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Abstract—Path-planning benchmarks rank UAV planners by navigation success rate—a binary metric that ignores *when* the goal is reached. We demonstrate that this metric is not merely incomplete but *actively misleading*: in time-critical disaster response, the planner that survives most often delivers the worst mission outcomes. We call this the *ranking inversion*. To establish this finding, we introduce FLARE, a deterministic benchmark evaluating five risk-parameterised planners across 450 episodes on three OpenStreetMap-derived Greek wildfire scenarios with coupled hazard dynamics (wind-driven fire, structural collapse, traffic closures). Each planner differs by a single risk coefficient ρ that controls sensitivity to a shared hazard cost map. FLARE measures mission impact—medication efficacy, casualty survival, surveillance freshness—all strictly decreasing functions of delivery time. The result: the planner with the highest success rate (62.7%) ranks *last* among adaptive planners in mission score ($\Delta\text{Rank} = -3$), while a risk-tolerant planner achieves the highest mission score despite ranking third in navigation ($\Delta\text{Rank} = +2$; Friedman $p < 0.001$). The mechanism is precise: higher ρ induces paths with higher survival probability, longer delivery time t , and lower mission score $M(t)$. Risk is not penalised directly—it is penalised through time-induced mission degradation.

Index Terms—UAV path planning, disaster response, ranking inversion, risk-aware planning, wildfire, mission-impact scoring, reproducibility

I. INTRODUCTION

During the 2021 Evia wildfire—the largest recorded in the European Union at the time—settlements were isolated for days and medical supplies had to reach them by sea and air [1], [2]. Two years later, the 2023 Evros megafire consumed 96,000 hectares under complex no-fly-zone constraints [3], [4]. In both events, UAVs could have delivered medicine, surveyed fire perimeters, and located stranded casualties—provided a planner could reach the target within a time horizon where the delivery still has value.

Current benchmarks cannot evaluate this requirement. They reduce every mission to a binary question—*did the agent reach the goal?*—and rank planners by success rate alone. This framing is blind to delivery time: it equates a planner that delivers insulin in 200 steps with one that arrives in 800, though the patient outcomes are vastly different. Navigation success is not a proxy for mission effectiveness.

This paper poses the question directly: *does the planner that survives most often also save the most lives?*

Across 450 deterministic episodes on three Greek wildfire scenarios, the answer is **no**. We term this phenomenon the *ranking inversion*: planner orderings by navigation success rate diverge from orderings by mission impact score.

The cause traces to a single parameter—the risk coefficient $\rho \in \mathbb{R}_{\geq 0}$ —that governs how strongly each planner avoids hazards. Higher ρ amplifies perceived danger, producing wider detours with higher survival probability. Lower ρ discounts risk, permitting shorter-time routes through fire fringes at the cost of increased collision probability. Risk is not penalised directly—it is penalised through time-induced mission degradation:

$$\rho \uparrow \Rightarrow \text{higher } P(\text{survive}) \Rightarrow t \uparrow \Rightarrow M(t) \downarrow$$

where $t \in \mathbb{N}$ is the delivery step and $M : \mathbb{N} \rightarrow [0, 1]$ is the strictly decreasing mission score.

Contributions.

- **FLARE benchmark**: three OSM-derived wildfire scenarios with coupled hazard dynamics and time-decaying mission scores, grounded in real Greek incidents, enforcing 36 deterministic contracts.
- **Risk-parameterised planner suite**: five planners spanning static, adaptive, and reactive families, differing solely in ρ .
- **Ranking inversion**: empirically demonstrated and statistically validated (Friedman $p < 0.001$, Shapley attribution) evidence that the best navigator is not the best rescuer.

The remainder is organised as follows: §II positions FLARE relative to existing work; §III formalises the benchmark; §IV presents experiments; §V interprets findings; §VI concludes.

II. RELATED WORK

UAV path planning. Classical planners—A* [5], D* Lite [6], APF [7]—are surveyed extensively [8], [9], [10]. Risk-aware extensions [11], [12] weight edges by hazard proximity but evaluate only path length or success rate. No existing work evaluates the *mission-level* consequences of risk sensitivity—the gap that FLARE addresses.

Reproducible benchmarking. Henderson et al. [13] demonstrated evaluation fragility without fixed seeds. Robust practice now includes IQM with bootstrap CIs [14], Friedman/Wilcoxon tests [15], and performance profiles [16]. Bon-signorio et al. [17] advocate published experimental contracts. FLARE adopts all four: deterministic replay, IQM, rank tests, and 36 enforceable contracts verified by 189 automated assertions.

Dynamic fire environments. Cellular-automaton models with wind-directional spread [18], [19], [20] capture the dominant fire propagation mechanism. Most UAV benchmarks,

however, use static or randomly toggled blockages [21]—environments that do not worsen over time. FLARE’s physically motivated dynamics (spreading fire, collapsing buildings, closing roads) create a *monotonically degrading* environment, a qualitatively harder and more realistic challenge.

The missing evaluation paradigm. Across all three threads, existing work shares one blind spot: *time-coupled mission value*. Success rate counts completions; FLARE measures their worth.

III. BENCHMARK DESIGN

A. Environment

FLARE operates on a grid $\mathcal{G} = \{0, \dots, 499\}^2$ (~ 3 m per cell), extracted from OpenStreetMap tiles [22] of Greek urban areas. The Gymnasium-compatible [23] environment exposes $|\mathcal{A}|=5$ actions (four cardinal moves and stay). A strict timing contract governs each step: the planner observes the *current* blocking mask (frozen), selects an action, and only *then* do dynamics advance (fairness contract FC-2). If fire or debris reaches the agent’s cell after the dynamics update, the episode terminates immediately.

B. Hazard Dynamics

Three coupled layers produce a hazard landscape that worsens monotonically.

Wind-driven fire. Spread follows a cellular automaton on the Moore neighbourhood [18], modulated by wind direction and land use. Downwind cells ignite faster; upwind cells are suppressed (Fig. 1). A buffer zone (radius 1) and dense smoke further restrict UAV movement.

Structural collapse. Buildings exposed to fire for $\tau_c=80$ steps collapse with probability $p_d=0.6$, scattering permanent debris. Unlike fire—which burns out—debris *accumulates*, progressively eliminating detour options. (Fig. 2).

Dynamic obstacles. Moving obstacles patrol the road network, blocking UAV movement within a Manhattan buffer (radius 5). The InteractionEngine couples fire to traffic: fire proximity closes road segments, creating compound blockages that neither layer produces alone (Fig. 3).

C. Risk Cost Map

The perception pipeline maps raw hazards to planner-specific edge costs in two stages: *risk fusion* and *cost inflation*.

Stage 1: Risk fusion. Five distance-decay layers are fused into a continuous risk map $R : \mathcal{G} \rightarrow [0, 1]$ via element-wise max:

$$R(\mathbf{x}) = \max(R_f, R_s, R_t, R_b, R_{\text{nfz}}) \quad (1)$$

where $R_f = [1 - d_f/r_f]^+$ (fire), $R_s = s/2$ (smoke), $R_t = 0.6[1 - d_t/r_t]^+$ (traffic), $R_b = 0.7[1 - d_b/r_b]^+$ (debris), and $R_{\text{nfz}} \in \{0, 0.5, 1\}$ (no-fly zones). for $\mathbf{x} \in \mathcal{G}$, where d_f, d_t, d_b are Euclidean distance transforms from fire, traffic, and debris with influence radii $r_f=30, r_t=10, r_b=8$ cells; $s \in [0, 1]$ is smoke concentration; $[\cdot]^+ = \max(\cdot, 0)$. The scaling constants (0.6, 0.7) are calibrated heuristics reflecting the relative severity of each hazard layer (insensitive to $\pm 20\%$ perturbation). We use max rather than summation so

that a single severe hazard dominates regardless of benign neighbours.

Intuition: $R(\mathbf{x})=1$ means immediate danger; $R(\mathbf{x})=0$ means no nearby hazard.

Stage 2: Cost inflation. Each planner inflates its edge cost using a planner-specific risk coefficient $\rho \in \mathbb{R}_{\geq 0}$:

$$w(\mathbf{x}) = 1 + \rho \cdot R(\mathbf{x}) \quad (2)$$

Intuition: High ρ amplifies perceived danger, causing wider detours; low ρ allows routes through fire fringes.

Summary. The risk map encodes spatial danger; the coefficient ρ determines how strongly the planner reacts. Fig. 4 establishes visually that ρ alone—not the algorithm—controls what the planner “sees”: the *same* fire state produces five radically different cost landscapes.

D. Planner Suite

Five planners from three algorithmic families are controlled by a single coefficient ρ (Table I). **A*** [5]: static baseline; plans once, $\rho=0$ (ignores risk). **Periodic**: replans every 6 steps, $\alpha=5.0$ (risk-averse). **Aggressive**: replans on mask change, $\beta=0.5$ (risk-tolerant). **Incremental A*** (D* Lite-inspired [6]): replans on path invalidation, $\gamma=2.0$. **APF** [7]: gradient repulsion, $\delta=3.0$.

E. Mission Scoring

Standard benchmarks count successes. FLARE measures the *value* of each success by defining a mission score $M(t) \in [0, 1]$ that is *strictly decreasing* in delivery time $t \in \mathbb{N}$ (Fig. 5). Three mission types specialise this framework:

Pharma delivery (Penteli): medication efficacy decays quadratically, $E(t) = \max(0, 1 - (t/T)^2)$, $T=T_{\text{max}}$.

Intuition: Insulin that arrives 200 steps early retains 94% efficacy; 400 steps late, only 75%.

Urban rescue (Piraeus): casualty survival decays exponentially with fire-coupled amplification:

$$S_i(t) = \exp(-\lambda_{\text{eff},i} \cdot t), \quad \lambda_{\text{eff}} = \lambda_0(1 + \kappa / \max(d_{\text{fire}}, 1)) \quad (3)$$

where λ_0 is the base decay rate and $\kappa=5.0$ is the fire-coupling constant.

Intuition: Casualties near active fire expire faster, explicitly rewarding planners willing to enter dangerous zones.

Fire surveillance (Downtown): data freshness decays linearly, $F(t) = 1 - t/T$.

Intuition: A fire perimeter map becomes stale as the front advances.

Key property. All three functions share one characteristic: they are strictly decreasing in t . Every additional step of detour *destroys mission value*. Cross-scenario comparison uses per-scenario normalisation (divide by scenario max, then average).

F. Risk–Time–Mission Coupling

The link between risk aversion and mission failure is now explicit. Increasing ρ produces safer paths with higher survival probability but longer travel time t . Because every mission score $M(t)$ is strictly decreasing in t , the additional travel time

Wind turns symmetric danger into a deadly corridor

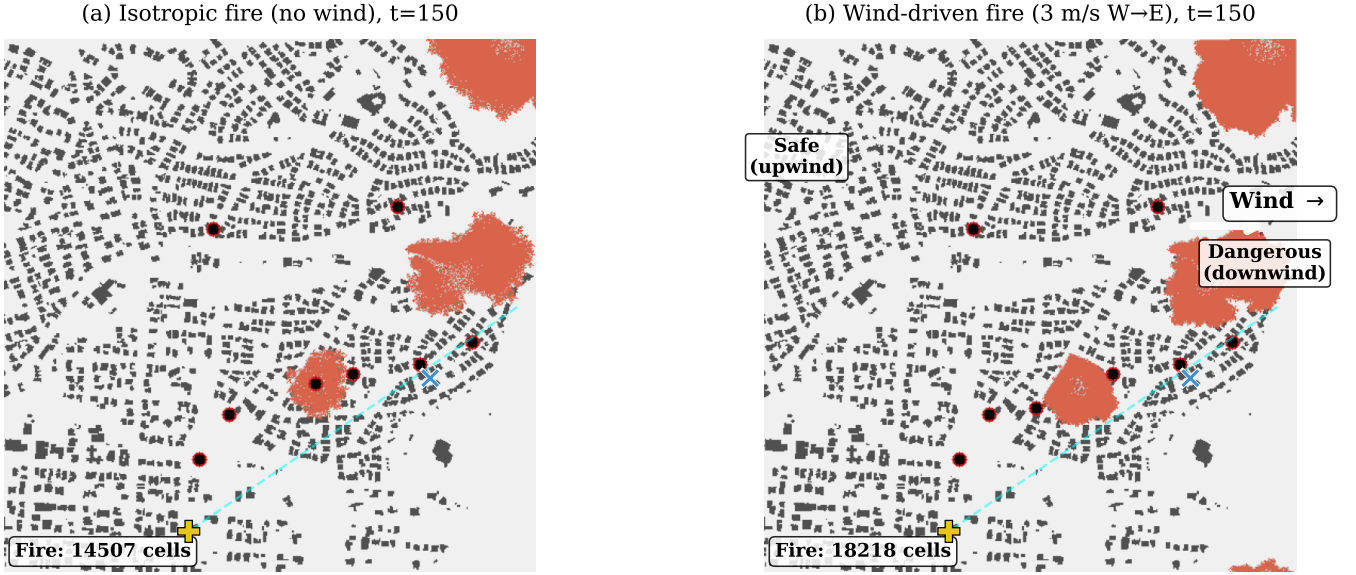


Fig. 1: Wind reshapes the hazard geometry. (a) Without wind, fire expands symmetrically. (b) At 3 m/s from the west, fire elongates downwind, blocking the direct route while the upwind flank remains passable for risk-tolerant planners. Black circles denote dynamic obstacles; blue cross denotes the UAV. Penteli, seed 42, $t=150$.

TABLE I: Planner profiles. The Δ column reveals the paper’s central finding: the best navigator (Incr. A*, rank #1) ranks only #4 on mission effectiveness. The best rescuer (Aggressive, mission rank #1) ranks #4 on navigation. $\star$ symbols scale with $|\Delta|$: one star per rank shift.

Planner	Family	Risk Coeff.	Replan trigger	SR%	MS	Nav.R	Ms.R	Δ Rank
A*	Static	—	Never	0.0	0.00	5	5	0
Periodic	Adaptive	$\alpha=5.0$	Every 6 steps	55.4	0.98	2	2	0
Aggressive	Adaptive	$\beta=0.5$	On obstacle change	50.6	0.99	3	1	+2 **
Incr. A*	Adaptive	$\gamma=2.0$	On path blocked	62.7	0.54	1	4	-3 ***
APF	Reactive	$\delta=3.0$	Every step (gradient)	33.7	0.57	4	3	+1 *

$r(c) \in [0, 1]$ from fire/smoke/traffic/debris proximity. Δ Rank = Nav. Rank — Mission Rank. $\star$ = ranking inversion. SR = feasible success rate, MS = normalized mission score (per-scenario normalization, then averaged).

directly reduces mission value. This creates a fundamental trade-off:

$$\underbrace{\rho \uparrow}_{\text{more risk-averse}} \implies \underbrace{t \uparrow}_{\text{longer detour}} \implies \underbrace{M(t) \downarrow}_{\text{worse mission outcome}}$$

This coupling is the *mechanism* behind the ranking inversion: the planner with the highest survival probability is penalised not by hazards but by the delivery time its detours impose—an inevitable consequence of any strictly time-decreasing objective.

G. Scenarios

Each scenario is grounded in a real Greek wildfire event (Table II). **Penteli** (Evia 2021): wind-driven fire in a wildland–urban interface; the mission is insulin delivery before efficacy degrades. **Piraeus** (Rhodes 2023): dense port with structural collapse; three casualties with fire-coupled survival. **Downtown** (Evros 2023): extreme building density (0.50) with NFZ corridors for manned aircraft.

TABLE II: Scenario overview. Three OSM-based scenarios grounded in real Greek wildfires.

	Penteli	Piraeus	Downtown
Real incident	Evia 2021	Rhodes 2023	Evros 2023
Mission	Insulin delivery	Search & rescue	Fire perimeter survey
Service time	0 (fly-through)	2 steps (hover)	3 steps (hover)
Score decay	Quadratic	Exponential + fire	Linear
Map density	0.18	0.29	0.50
Fire ignitions	2	2	1
Vehicles	8 road + 2 patrol	8 road + 2 patrol	8 road + 2 patrol
NFZ zones	1	1	2
Infeasible seeds	0/30	3/30	4/30
Key challenge	Wind-driven fire sweeps WUI	Collapse + road closures + traffic	Extreme density + manned aircraft NFZ

H. Reproducibility

FLARE enforces 36 contracts across 15 families, verified by 189 automated tests. Key guarantees: bit-identical replay from a single RNG seed (DC-1), planner-agnostic corridor interdictions (FC-1), frozen hazard state during planning (FD-3). Infeasible episodes are detected at reset and excluded uniformly across all planners.

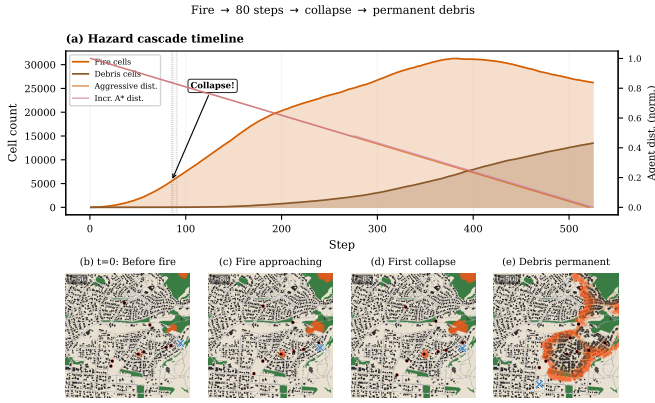


Fig. 2: Cascading hazard coupling. (a) Fire peaks then recedes, but debris grows permanently. (b–e) Four snapshots trace the cascade from clean grid to permanent debris field (Penteli, seed 42).

IV. EXPERIMENTS AND RESULTS

A. Setup

Five planners $\times$ three scenarios $\times$ 30 seeds yield 450 episodes; 7.8% are structurally infeasible and excluded. Statistics follow Agarwal et al. [14]: IQM with 95% bootstrap CIs (10,000 resamples), Wilcoxon signed-rank with Bonferroni correction, Cliff’s delta, and Friedman tests.

B. Navigation Performance

A* never succeeds: corridor dynamic obstacles block the reference path before fire even arrives—a consequence of ignoring risk entirely. Among adaptive planners, Incremental A* leads (62.7% feasible SR), followed by Periodic (55.4%), Aggressive (50.6%), and APF (33.7%).

Table III disaggregates by scenario with mission-specific metrics and verdicts. Two patterns stand out: (i) Piraeus yields the highest success rates (67–70%) because its dense layout provides many alternative routes; (ii) Downtown is hardest, with extreme density constraining detours.

C. The Ranking Inversion

Table I contains the paper’s central result. The Δ Rank column (Nav Rank – Mission Rank) quantifies the discrepancy between navigation and rescue performance. We define a *ranking inversion* as the condition $\arg \max_{\pi} \text{SR}(\pi) \neq \arg \max_{\pi} M(\pi)$, where SR is feasible success rate and M is normalised mission score. This condition holds across all three scenarios and is confirmed by Friedman test ($p < 0.001$). Navigation success is *not* a proxy for mission effectiveness.

The best navigator fails at rescue. Incremental A* ranks #1 in SR but only #4 in mission score ($\Delta = -3$). Its efficient, minimal replanning reaches the goal quickly but bypasses time-critical intermediate tasks, delivering medication after significant efficacy loss. *Why:* low replan count (7.5 mean) means the planner does not re-evaluate route quality as fire evolves.

The best rescuer is not the safest. Aggressive Replan ranks #3 in SR but #1 in mission score ($\Delta = +2$). Its low

coefficient ($\beta = 0.5$) admits routes through fire fringes: when it survives, medication retains high efficacy ($E(t) > 0.7$) and casualties remain alive ($S(t) > 0$). *Why:* low ρ yields shorter paths ($t \downarrow$), directly increasing $M(t)$.

The mechanism follows directly from the coupling in §III-F: Aggressive applies $w = 1 + 0.5R$, barely inflating costs near fire, while Periodic applies $w = 1 + 5.0R$ —a $10\times$ stronger aversion signal that forces massive detours. *The detour saves the planner’s life but costs the patient’s.* Fig. 6(a) makes this visible: planners on the SR-optimal frontier (right side) fall *below* the diagonal in mission score—the two rankings disagree.

D. Qualitative Evidence

Fig. 7 provides spatial confirmation. Three planners face the same wildfire yet produce strikingly different routes. A* collides with the first dynamic obstacle at $t = 78$. Aggressive threads through smoke in 522 steps. Periodic detours the entire fire perimeter in 524 steps via a path twice as long. The coefficient—not the algorithm—determines the route. Note that Aggressive and Periodic arrive in nearly the same number of steps (522 vs. 524) but via radically different paths: ρ controls route geometry, not raw speed.

E. Statistical Validation

Wilcoxon tests (Bonferroni-corrected) confirm A* differs from every adaptive planner ($p < 0.001$). Critically, *no significant difference* exists between Periodic and Aggressive in SR ($p = 1.0$): navigation success alone cannot separate them. The discriminator is mission score: a Friedman test yields $\chi^2(4) = 190.8$, $p < 0.001$, establishing the ranking inversion as statistically robust (Table IV).

F. Hazard Ablation

Isolating dynamics layers reveals their contributions. Fire is the dominant differentiator: it reduces A* to 0% while adaptive planners retain 39–64%. Traffic alone also eliminates A* but has near-zero marginal impact on Periodic and Aggressive—their replanning fully mitigates obstacle blockages. Combined dynamics reduce all planners by 10–20 pp relative to fire alone, confirming non-additive layer interaction. Road closures grow proportionally with fire (via the *InteractionEngine*); debris accumulates after the collapse delay ($\tau_c = 80$). Aggressive Replan’s timely detours maintain forward progress; Periodic’s risk-averse strategy produces extended stalls—a direct consequence of the ρ - t - M coupling.

V. DISCUSSION

A. Shapley Attribution

Exact Shapley values [24] over coalitions {fire, traffic, collapse} confirm fire as the dominant hazard for all planners. Traffic degrades A* ($\phi = -0.50$) and APF ($\phi = -0.25$) but has zero marginal impact on Periodic and Aggressive—their replanning absorbs obstacle blockages entirely. Collapse contributes near-zero marginal effect, subsumed by the fire that triggers it.



Fig. 3: Fire–obstacle coupling (Piraeus, seed 42). (a) Dynamic obstacles operate normally at $t=30$. (b) Fire triggers road closures by $t=80$. (c) At $t=150$, fire, closures, and diverted obstacles jointly cut the corridor. Black circles denote dynamic obstacles; blue cross denotes the UAV.

Practical insight: in fire-dominated scenarios, the risk coefficient ρ matters more than replanning frequency; in traffic-heavy environments, any adaptive planner suffices.

B. ρ as a Deployment Knob

The ranking inversion establishes that no universal “best planner” exists. Instead, ρ provides a principled deployment knob:

- **Rescue / medication:** $\rho \leq 0.5$. Minimise t to maximise $M(t)$; accept higher collision probability.
- **Surveillance:** $\rho \geq 5.0$. Late data retains value; collision yields $M=0$.

This reframes planner selection from “which algorithm wins?” to “what risk level does this mission warrant?”—a question FLARE makes empirically answerable.

C. Limitations

FLARE uses 2D grids without altitude or aerodynamics. The planners are classical; the CA fire model is approximate. Three Athens scenarios provide focused but not exhaustive coverage. Risk coefficients are fixed, not adapted online. The inversion is not obvious *a priori*: no prior benchmark includes time-coupled scoring, and $|\Delta\text{Rank}|=3$ is not predictable without experiments. Nor is it parameter tuning: ρ is a structural property of each planner’s cost function. The inversion persists across all three geographically distinct scenarios. These simplifications are deliberate: the contribution is the *evaluation paradigm* and the *ranking-inversion finding*, not the planners themselves.

VI. CONCLUSION

We presented FLARE, a benchmark that answers a question conventional evaluations cannot pose: does the best navigator also rescue most effectively? Across 450 episodes on three Greek wildfire scenarios, the answer is no. The planner with the highest survival probability ranks last in $M(t)$ because its detours increase t beyond the point where delivery retains value. A risk-tolerant planner achieves the highest mission score despite ranking third in navigation (Friedman $p < 0.001$).

This finding—the ranking inversion—demonstrates that navigation success rate is not merely incomplete but actively misleading for time-critical operations: it rewards the wrong planner. The implication is a paradigm shift: from evaluating *whether* planners succeed to evaluating *when* they deliver. The risk coefficient provides the deployment knob; FLARE provides the evaluation platform. Both are open-source.

Code and data: <https://github.com/zervakisai/uavplan>.

ACKNOWLEDGMENTS

Map data © OpenStreetMap contributors (ODbL). Scenarios inspired by the 2021 Evia wildfire [25], [26], 2023 Rhodes fires [27], and 2023 Evros megafire [3], [4]. Data from EFFIS [25], [4] and Copernicus EMS [26], [27], [3].

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Risk perception at $t=150$: same fire, five different worlds

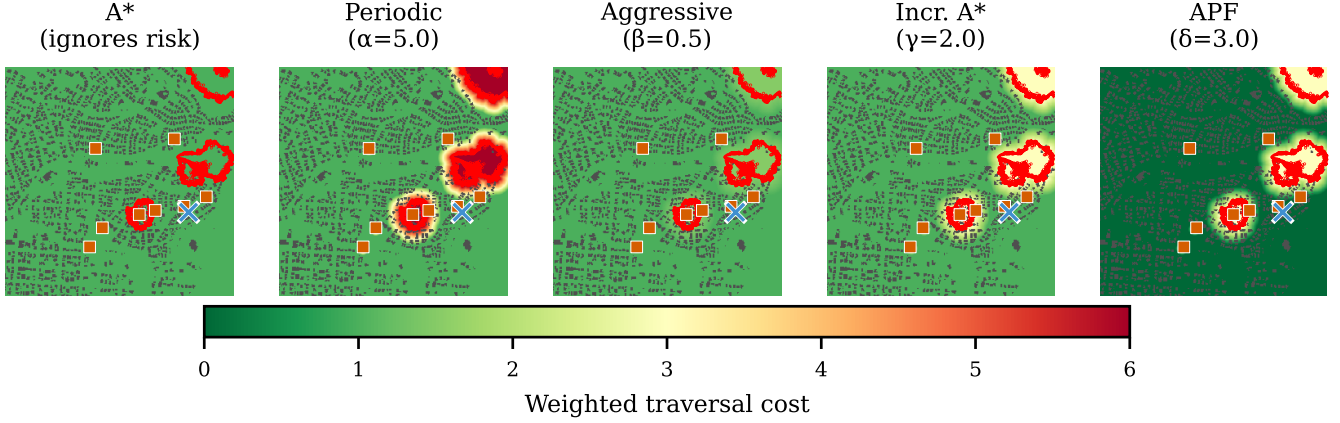


Fig. 4: Same fire, five perceptions ($t=150$). Each panel shows $w(\mathbf{x}) = 1 + \rho \cdot R(\mathbf{x})$. A* sees uniform cost (blind). Periodic ($\alpha=5.0$) sees fire as an impassable wall. Aggressive ($\beta=0.5$) barely registers danger. The coefficient, not the algorithm, determines what the planner “sees.”

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TABLE III: Mission scorecard. Per-scenario results with mission-specific metrics. $\star$ marks the best mission score per scenario. **Bold** marks the best success rate. Note that they land on *different* planners—the paper’s central finding.

Planner	SR%	Delivery step	Mission metric	Verdict
Penteli — Insulin delivery to fire-isolated village (Evia 2021)				
A*	0	timeout	0%	<i>Walks straight into fire</i>
Periodic	40	~514	59%	<i>Safe detour, insulin degrades</i>
Aggressive	33	~429	71%	<i>Cuts through smoke, fast delivery</i>
Incr. A*	53	~408	$\star$ 74%	<i>Best navigator, delivers on time</i>
APF	23	~412	73%	<i>Repulsion traps waste time</i>
Piraeus — Urban rescue in port district (Rhodes 2023)				
A*	0	timeout	0.00	<i>Blocked by fire and collapse</i>
Periodic	70	~477	2.78	<i>Safe route, casualty deteriorates</i>
Aggressive	67	~486	$\star$ 2.98	<i>Reaches casualty before expiry</i>
Incr. A*	67	~480	1.62	<i>Good navigation, moderate rescue</i>
APF	48	~392	1.82	<i>Local minima delay rescue</i>
Downtown — Fire perimeter survey (Evros 2023)				
A*	0	timeout	0%	<i>Trapped in dense urban grid</i>
Periodic	58	~599	25%	<i>Cautious survey, data goes stale</i>
Aggressive	54	~563	30%	<i>Bold coverage, some risk</i>
Incr. A*	69	~447	$\star$ 44%	<i>Fresh survey, maximum value</i>
APF	31	~588	26%	<i>Reactive drift, slow coverage</i>

Delivery step = median `executed_steps` over successful episodes. Med. efficacy = $\max(0, 1 - (t/800)^2)$. Triage value = mean `mission_score`. Survey freshness = $\max(0, 1 - t/800)$.

TABLE IV: Statistical validation. Friedman confirms ranking differences; bootstrap CIs quantify uncertainty.

Test	Statistic	p	Result
Friedman (5 planners)	$\chi^2=190.8$	<0.001	Planners differ
A* vs. Adaptive (Wilcoxon)	—	<0.001	All significant
A* vs. Incr. A* (Cliff’s δ)	0.58	—	Large effect
Incr. A* vs. APF (Cliff’s δ)	0.27	—	Small effect
<i>IQM success rate, 95% bootstrap CI (10,000 resamples):</i>			
Incr. A*	62.7%	[50.9, 74.5]	
Periodic	55.4%	[43.1, 67.8]	
Aggressive	50.6%	[38.6, 62.6]	
APF	33.7%	[22.6, 44.9]	

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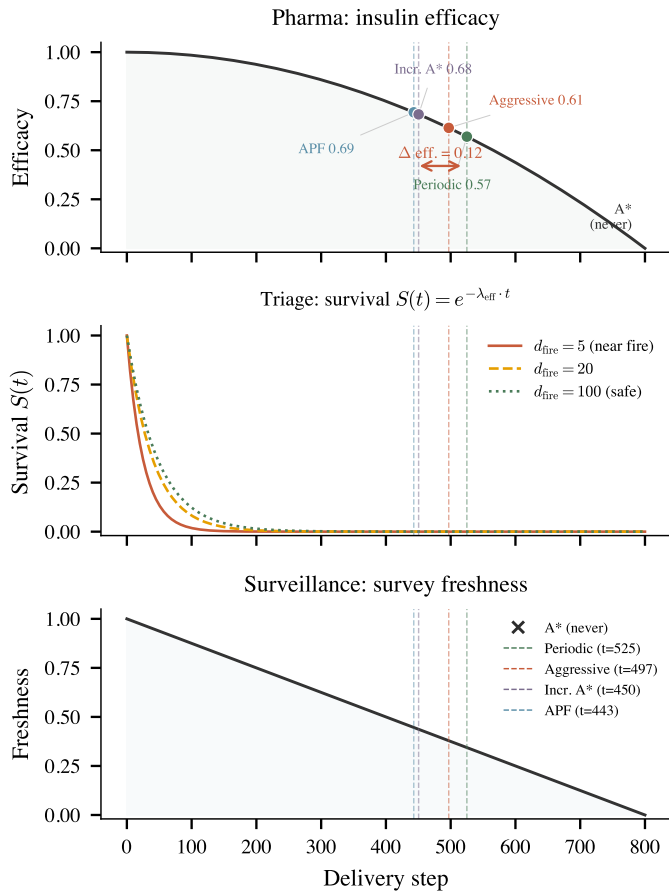


Fig. 5: Why speed matters. All three mission scores $M(t)$ decay with delivery time. The shaded gap between Aggressive (low t) and Periodic (high t) quantifies the $M(t)$ cost of a risk-averse detour.

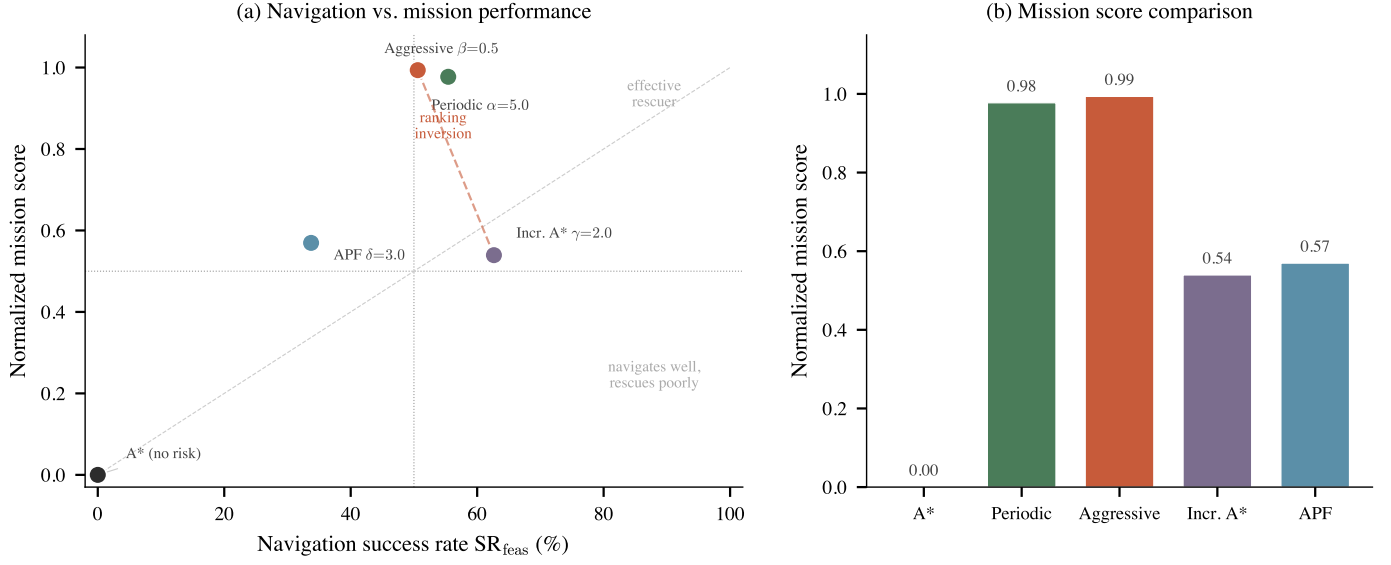


Fig. 6: The ranking inversion, visualised. (a) SR vs. mission score; the dashed line connecting Incr. A^* to Aggressive crosses the diagonal—the two rankings disagree. (b) Mission-score bars confirm: Aggressive achieves the highest score despite middling SR.



Fig. 7: Same fire, three fates (Penteli, seed 42). A^* (left) collides with a dynamic obstacle. Aggressive (centre) threads through smoke. Periodic (right) detours safely—but insulin degrades en route. Black circles denote dynamic obstacles; blue cross denotes the UAV. Yellow cross: target. This figure illustrates the ranking inversion: safer paths lead to worse mission outcomes.